$$P^A = \sum_{i=1}^{N} \frac{\beta \pi_i u_j'(c_j^i)}{u_j'(c_j^0)} x_i = \sum_{i=1}^{N} \pi_i \left(\frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)} x_i \right) = E\left[\frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)} x_i \right]$$

$$1 = E\left[\frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)} \frac{x_i}{P^A} \right] \qquad R_i = \frac{x_i}{P^A} = Gross\ return$$

$$1 = E\left[\frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)} R_i \right] \qquad m = \frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)} \quad stochastic\ discount\ factor$$

This can be used to price any assets

For risk free asset:

$$1 = E[m * R_f]$$

$$1 = R_f\, E[m]$$

$$E[m] = \frac{1}{R_f}$$

For random stock k:

$$1 = E[m * R^k] = E[m]E[R^k] + Cov(m, R^k)$$

$$1 = \frac{1}{R_f} E[R^k] + Cov(m * R^k)$$

$$E[R^k] - R_f = -R_f\, Cov(m, R^k) \qquad ①$$

$$E[XY] = E\big[(X - E(X) + E(X))(Y - E(Y) + E(Y))\big]$$

$$= E\big[(X - E(X))(Y - E(Y))\big] + \underbrace{E[X - E(X)]E[Y]}_{=0} + \underbrace{E[X]E\big[Y - E[Y]\big]}_{=0} + E[X]E[Y]$$

$$= E[X]E[Y] + Cov(X, Y)$$

For $Cov(m, R^k)$ to be high → m & R^k move in the same direction

$m = \frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)}$ → R^k & c_j^i move in opposite direction.

If R^k positively comove with c_j^i → R^k negatively comove with $u_j'(c_j^i)$ → $E[R^k]$ is high

When consumption increase (good state), pay money; when consumption decrease (bad state), pay no money → can't provide insurance → require higher expected return

When consumption and R^k are negatively co-move, Asset k is similar to insurance. Demand for this type of asset is high, driving up its price and resulting in a low expected return.

For market portfolio:

$1 = E[m * R^m] = E[m]E[R^m] + Cov(m, R^m)$

$E[R^m] - R_f = -R_f Cov(m, R^m)$ ②

$E(R^k) - R_f = -R_f\ Cov(m, R^k)$ ①

①/②

$\frac{E(R^k) - R_f}{E[R^m] - R_f} = \frac{Cov(m, R^k)}{Cov(m, R^m)}$ $\rightarrow become\ CAPM: E[R^k] - R_f = \beta\left(E(R^m) - R_f\right)$

$if\ m = R^m + constant$ $\frac{Cov(m, R^k)}{Cov(m, R^m)} = \beta$ $\rightarrow CAPM\ beta$

The difference between the A-D model and the CAPM

$In\ CAPM, m = R^m + constant$ systemic risk comove with market return.

$In\ AD\ model, m = \frac{\beta u_j'(c_j^i)}{u_j'(c_j^0)}$ systemic risk comove with aggregate consumption.